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An all-pairs catalogue of real three-dimensional close encounters between numbered main-belt asteroids during the Gaia DR3 window, with a measured incompleteness budget

D. Frascarelli[1]

Affiliation — TODO e-mail: dsanfra@gmail.com

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ABSTRACT

Context. Close approaches between asteroids are the classical route to asteroid mass determination and a dynamical observable of the main belt. Systematic encounter searches exist — from early surveys (Galád & Gray 2002) to mutual-encounter catalogues built for Gaia astrometry (Ivantsov et al. 2018) and the signal-ranked search behind the largest mass catalogue to date (Fuentes-Muñoz et al. 2025) — but their event lists are published truncated (top candidates per perturber) or without a characterised selection function: no all-pairs 3D encounter catalogue has been published in full with provenance and a *measured* incompleteness budget.

Aims. We build a systematic catalogue of real 3D close encounters between numbered main-belt asteroids (semi-major axis 1.5–4.0 AU) during the Gaia DR3 window (2014 July 25 – 2017 May 28), publish it in full, and, unlike a bulk propagation exercise, we *measure* what the detection pipeline misses rather than asserting completeness.

Methods. Orbits are propagated from a single frozen MPCORB snapshot centred on the window. Encounters are found by a per-timestep 3D KD-tree spatial query and refined on a dense temporal sub-grid. Reported distances use an analytic two-body refiner; a hybrid variant carries full N-body (rebound/ASSIST) refinement for the dynamically fragile (low-perihelion or high-eccentricity) subset. Perturber masses come from a joint mass+orbit least-squares fit to Gaia FPR astrometry.

Results. The catalogue records the minimum physical 3D separation — a genuine spatial approach, not a sky-plane co-location — for 72 236 904 candidate pairs within 0.05 AU, with encounter epoch, relative velocity, geometry, and estimated physical properties. Its distinguishing feature is a *measured* completeness budget: the Kepler-to-N-body refinement error (median 12 μAU, p99 2.5 mAU) and the threshold-induced false-negative rate (0.70 %, symmetric and scatter-dominated) bound this catalogue directly, while the recall of an orbital prefilter on the adverse tail (76 %) is quantified as a caveat for prefiltered runs. Applied to mass determination, the engine recovers all four calibrator masses (Ceres, Vesta, Pallas, Hygiea) at $|z| < 3$ and determines sixteen perturbers, of which ten identifiable masses agree with recent independent work.

Conclusions. The catalogue is a reusable, defensibly characterised target-selection resource for asteroid dynamics and mass determination. It is explicitly not complete; the large–large ($D \gtrsim 100$ km, measured) approaches it contains all involve well-studied bodies, and no unpublished large-body mass emerged from the numbered population.

Key words. minor planets, asteroids: general – celestial mechanics – astrometry – catalogs – surveys – methods: data analysis

1. Introduction

Close encounters between asteroids are of interest on several fronts. A slow, close approach between a massive perturber and a small test body deflects the latter's orbit measurably, and such events are the classical route to asteroid mass determination (Michalak 2000; Fienga et al. 2003; Goffin 2014; Baer & Chesley 2017; Siltala & Granvik 2020; Li et al. 2023; Fuentes-Muñoz et al. 2025). Encounters between members of the same dynamical family probe the collisional history of the belt, and the statistical distribution of mutual approach distances and velocities is itself a dynamical observable.

Systematic encounter searches have a long history. Galád & Gray (2002) searched for mass-determination encounters across 24 599 asteroids; Ivantsov et al. (2018) built a catalogue of mutual encounters of the numbered population over 2013–2023 specifically to exploit Gaia astrometry; Goffin (2014) fitted masses against the entire numbered population rather than curated target lists; and Fuentes-Muñoz et al. (2025) ran a systematic, signal-ranked search over 1 783 perturbers against more than a million orbits — using, for main-belt targets, the same 0.05 AU threshold adopted here. What none of these works provides is the *event list published in full with a characterised selection function*: published tables are truncated to the top candidates per perturber, and the completeness of the underlying searches is asserted rather than measured. That is the gap this work fills: an *all-pairs* catalogue over the numbered main-belt population ($1.5 \leq a \leq 4.0$ AU) in the Gaia DR3 window, published in full with provenance, where — the point that separates this catalogue from a bulk propagation exercise — we *measure* what the detection pipeline misses rather than asserting completeness.

A methodological distinction is central. Two asteroids can appear arbitrarily close in right ascension and declination as seen from Earth while remaining astronomical units apart along the line of sight. This catalogue detects the former case only when it is also the latter: it records the *minimum physical separation in three dimensions* between the two bodies' heliocentric positions, independent of any observer. It is therefore a catalogue of genuine spatial approaches, not of sky-plane co-localisations. Light-time correction — essential when comparing model positions to Gaia's astrometry — is deliberately *not* applied here, because the encounter is a purely geometric relation between two propagated trajectories at a common dynamical time.

The remainder of this paper describes the data and detection method (Sect. 2), the measured completeness budget that is the catalogue's distinguishing feature (Sect. 3), the notable individual events surfaced by mining the catalogue (Sect. 4), the application to mass determination (Sect. 5), data availability (Sect. 6), and conclusions (Sect. 7).

2. Data and method

2.1. Sources and conventions

Orbital elements are taken from a single frozen MPCORB snapshot, `MPCORB_20160217.DAT` (SHA-256 prefix `3e44e7d3...`), selected because its osculating epoch lies at the centre of the Gaia DR3 window, minimising the propagation distance — and hence the accumulated two-body error — for Kepler refinement. The snapshot hash and download date are recorded in the catalogue's provenance sidecar so that results are traceable to an exact input; MPCORB is a living file and results are not bit-reproducible across snapshot versions. The run is restricted to *numbered asteroids* (higher orbital quality); provisional designations are out of scope for this release. The sample is further cut to the main belt by semi-major axis, $1.5 \leq a \leq 4.0$ AU, which excludes near-Earth asteroids ($a < 1.5$ AU), the Jupiter Trojans, and the outer/trans-Jovian populations ($a > 4$ AU). The resulting universe is $N = 449{,}454$ propagated bodies; this N is the population denominator for every completeness and censoring figure below, and is recorded in the provenance sidecar. "Complete" therefore always means complete *with respect to this numbered main-belt sample*, never the whole catalogue of known asteroids.

The temporal domain brackets the Gaia DR3 Solar System survey period (Tanga et al. 2023): the propagation window is 2014 July 25 – 2017 May 28, whose start (the DR3 astrometric-solution baseline) slightly precedes the 2014 August 5 start of nominal SSO operations, so the window fully contains the survey coverage. All internal computation is carried out in Julian Date on the Barycentric Dynamical Time (TDB) scale; conversions to/from the Barycentric Coordinate Time (TCB) that Gaia reports, and to UTC, are performed only at input/output interfaces. Distances are in AU, angles in radians, and velocities in AU/day internally, converted to user-facing units (km, degrees, km s^{-1}) only for output. Positions are computed in the barycentric ecliptic frame for the N-body coarse scan and in the heliocentric ecliptic frame for the Kepler refinement; because the catalogue records the *mutual* separation of two bodies, the common Sun–barycentre offset cancels in the difference vector and the recorded distances are identical in either frame. (The provenance sidecar's `reference_frame` field refers to the N-body scan.)

2.2. Detection

The naive cost is prohibitive: ~150 000 bodies over a multi-year window at hour-level cadence implies of order 10^{10} pairwise distances per step. Detection is therefore layered.

An *orbital prefilter* is available to prune pairs that cannot physically approach — the shipped heuristic keeps a pair only if $|\Delta a| \leq 0.5$ AU and $|\Delta i| \leq 30°$. This filter is a deterministic mask on osculating elements. We note here, and quantify in Sect. 3.3, that at main-belt scale ($\gg$ 5000 bodies) the pair-list prefilter is not built; the frozen run therefore applied only the KD-tree spatial query, and the $|\Delta a|$ cut did *not* shape this catalogue (provenance `prefilter.effective = "skipped_large_n"`).

The core of the detector is a *per-timestep 3D KD-tree*. At each coarse step the heliocentric ecliptic positions of all bodies are inserted into a KD-tree ($O(N \log N)$ construction), and a radius query returns neighbour candidates ($O(\log N)$ per query), avoiding any $O(N^2)$ distance matrix. The coarse cadence is $\Delta t = 12$ h. To prevent the cadence from missing an encounter whose true minimum falls *between* two samples, the query radius is *widened* to $r_q = 0.05\,\mathrm{AU} + v_{\max}\,\Delta t \approx 0.0572$ AU with $v_{\max} = 25\,\mathrm{km\,s^{-1}}$: under near-linear relative motion a pair whose separation dips below 0.05 AU between samples is within r_q at the nearer bracketing sample and therefore enters the candidate set; the sub-grid refinement then discards any whose true minimum exceeds 0.05 AU. Bracketing a minimum midway between two samples strictly requires only $v_{\max}\,\Delta t/2 \approx 0.0536$ AU; the pipeline uses the full-step widening ($v_{\max}\,\Delta t$), a factor-of-two conservative margin. The bracketing is thus exact for relative velocities up to $v_{\max}$, and the residual (encounters with $v_{\rm rel} > v_{\max}$) is bounded in Sect. 3.3.

Each coarse candidate is then *refined on a dense temporal sub-grid*: a ±2 h window around the apparent minimum is sampled at 120 s, and the minimum separation, encounter epoch, and relative velocity are extracted from the refined track. Time blocks are independent, so the scan parallelises across processes.

Propagation is two-tiered. The coarse KD-tree scan is driven by an N-body rebound integration (WHFast; Sun + Jupiter + Saturn; $dt = 1$ h). The sub-grid refinement that produces the *reported* minimum distance runs through an analytic two-body Kepler propagator. Over the short Gaia DR3 arc (~3 yr from the central epoch) two-body propagation is adequate for the bulk of the belt, and its residual error is measured, not assumed (Sect. 3.1). For the dynamically fragile subset — pairs with perihelion $q_{\min} < 1.8$ AU or eccentricity $e_{\max} > 0.3$, 8 728 509 pairs (12.08 % of the catalogue) — a *hybrid* catalogue replaces the Kepler distances with full N-body refinement (rebound IAS15; Sun + 8 planets + 4 major asteroids; ±12 h around the Kepler minimum; 60 s sampling). The hybrid catalogue carries refinement_method ∈ {kepler, nbody} per row together with the Kepler and N-body values and their residuals, so users needing sub-mAU geometry on high-e / NEA-crossing pairs can cite it directly. Universal N-body refinement of all 72 M pairs (~400 CPU-days) is explicitly out of scope: the median error outside the fragile subset is already below 1 mAU (Sect. 3.1), so the marginal gain does not justify the cost.

2.3. Characterisation

Each pair contributes a single catalogue row: the *global* minimum separation of that pair over the observation window, with its exact epoch (from the refined sub-grid), the relative velocity at closest approach, and the geometry relative to Gaia. Secondary local minima of a pair that recurs within the window are not catalogued separately. A body is flagged *Gaia-observable* at the encounter when its solar elongation exceeds 45° (outside Gaia's solar-exclusion zone) *and* its apparent V magnitude is brighter than 21 (the faint limit); the pair-level flag is the logical OR of the two bodies. Of the 72 236 904 encounters, 13 640 870 (18.9 %) satisfy this for at least one body.

Physical properties are attached to both bodies. Diameters use direct measurements (IRAS/AKARI/NEOWISE/occultations, via the JPL Small-Body Database) where available; otherwise they are derived from the absolute magnitude H with an albedo taken — in decreasing order of preference — from a measured albedo, an orbital-zone average, or a default 0.14. A per-row provenance column records which of these applied, so *H*-derived (order-of-magnitude) sizes are distinguishable from measured ones in any size-based selection. The accompanying *class* is the *dynamical* class from the orbital elements (NEA/MBA/Trojan/Centaur/TNO), not a taxonomic (spectral) type.

The two encountering bodies are treated as independent test particles moving in the planetary + massive-asteroid field; their *mutual* gravitational deflection over the approach is not modelled. At asteroid masses and the $\gtrsim 10^5$ km separations that dominate the catalogue this is entirely negligible for the recorded geometry, and it is in any case the very quantity a downstream mass determination sets out to measure, not a property of the encounter itself.

2.4. Catalogue overview

The parent population and the encounter statistics are summarised in Figs. 1–3. The minimum-separation distribution (Fig. 1) rises monotonically toward the 0.05 AU threshold, as expected for a roughly uniform spatial density of encounter geometries; the relative-velocity distribution (Fig. 2) peaks near its median of $4.25\,\mathrm{km\,s^{-1}}$, typical of main-belt crossing velocities. The orbital distribution of the 93 010 encountering bodies (Fig. 3) reproduces the belt's known structure — the 3:1, 5:2 and 2:1 Kirkwood gaps appear as clear voids — confirming the sample is the ordinary numbered population, not a biased subset.

Ranking by mass-perturbation signal. Sheer row count is not utility: the overwhelming majority of the catalogued pairs are sub-km × sub-km approaches with no measurable dynamical signal. To let users select the subset that matters — as signal-ranked searches such as Ivantsov et al. (2018) and Fuentes-Muñoz et al. (2025) do — each row carries a deflection-signal estimate, the impulse-approximation velocity kick $\Delta v \simeq 2GM_1/(b\,v_{\rm rel})$ imparted by the larger body (`deflection_dv_m_s`; GM_1 from the measured diameter and a zone-average density). It is a ranking proxy, not a measurement. Ranked by it, the catalogue is steeply top-heavy: only 781 pairs exceed $1\,\mathrm{mm\,s^{-1}}$ and ∼4 000 exceed $0.1\,\mathrm{mm\,s^{-1}}$, and the top of the list is dominated by (1) Ceres and (4) Vesta in slow, close passes — precisely the classical mass-determination configuration. This column turns the 80 M-row bulk into a targetable resource; the mass-determination application of Sect. 5 draws its test bodies from its high-signal tail.

3. Completeness budget

The value of this catalogue is not that it detects encounters — that is a bulk propagation — but that it states, with measured numbers, what it *misses* and how large the Kepler-vs-N-body distortion is. We stress the scope: the three *measured* terms below (Kepler-vs-N-body refinement error, threshold censoring, prefilter recall) all compare the *same* input elements under different force models or cuts, so they quantify *pipeline-induced* incompleteness, not the physical completeness of the true encounter set. A fourth term — the uncertainty of the input osculating elements themselves — is *not* propagated here and is treated as a bound: for numbered main-belt asteroids the along-track error grows with $|t - t_{\rm epoch}|$ and, over the ± 1.4 yr window, ranges from sub-km for long-arc bodies to 10^3–10^4 km for short-arc, faint, high-number ones. It does not affect the completeness *fractions* below (those are pipeline-internal) but it does bound the physical significance of individual tabulated distances (Sect. 4.2, Sect. 4.3). A full propagation of AstDyS/JPL element covariances is deferred to a future release. Each measured term is reproducible from a dedicated script; a synthetic injection–recovery test (minima placed uniformly between coarse samples, high $v_{\rm rel}$ and e) is a permanent regression guard on the detection layer.

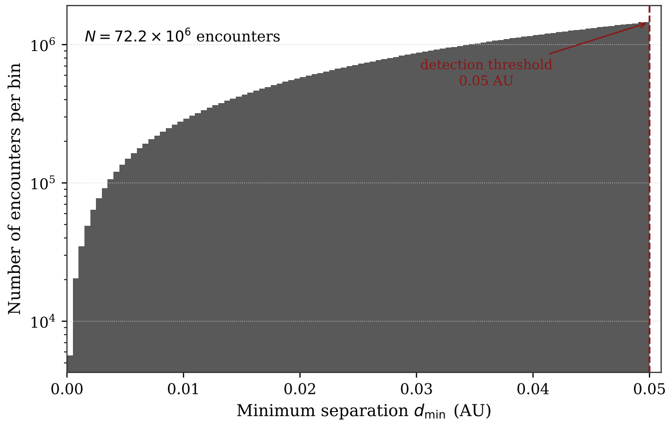

Fig. 1. Distribution of minimum 3D separations for the 72.2×10^6 catalogued encounters (log count per bin). The dashed line marks the 0.05 AU detection threshold.

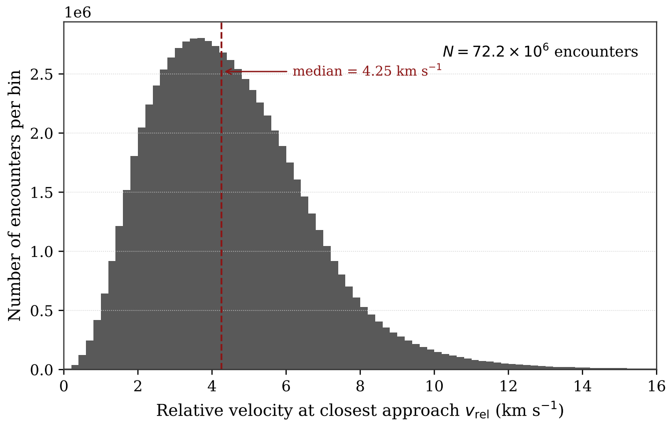

Fig. 2. Distribution of relative velocity at closest approach; the median is $4.25\,\mathrm{km\,s^{-1}}$.

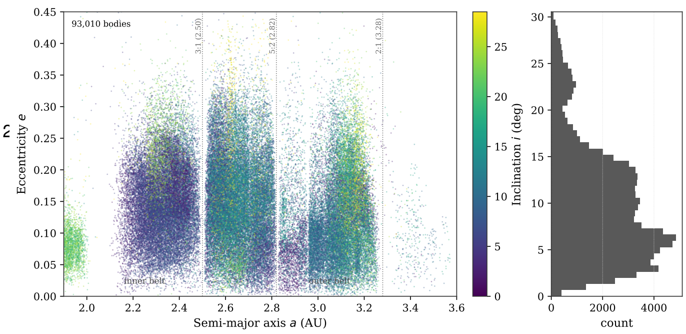

Fig. 3. Semimajor axis vs eccentricity of the 93 010 encountering bodies, coloured by inclination, with the inclination marginal at right. The 3:1, 5:2 and 2:1 mean-motion resonances (Kirkwood gaps) are marked.

3.1. Kepler vs N-body refinement error

The reported distances in the frozen catalogue come from the two-body refiner. We measured the resulting error against full N-body refinement in two stages. In *Stage A* (964 stratified pairs re-refined under full N-body) the median $|\Delta d|$ = 12 μAU, p95 = 678 μAU, p99 = 2.5 mAU, max = 11.3 mAU. In *Stage B* (production N-body refinement of the entire 8.73 M fragile subset) p99 $|\Delta d|$ = 1.99 mAU, max = 15.2 mAU, with 0 failed integrations, 0 unconverged, and maximum energy drift 5.6×10^{-14}.

For scale, 1 mAU $\approx 1.5 \times 10^5$ km, and the detection threshold is 0.05 AU. The median distortion is thus ~4 orders of magnitude below the threshold, and even the Stage B worst case is ~30 % of a threshold — confined to the fragile subset, which the hybrid catalogue refines. A separate check of the truncated perturber set (the scan used only Sun + Jupiter + Saturn) shows that adding the remaining planets shifts closest-approach distances by a median 1.3 μAU (p99 67 μAU, max 80 μAU), i.e. ~100× below the Kepler two-body error — the perturber truncation is not the dominant error term.

3.2. Threshold censoring (measured false-negative rate)

Because the catalogue contains only pairs whose *Kepler* distance is below 0.05 AU, a subtle bias arises at the threshold. When the fragile subset is re-refined to N-body, 25 283 pairs (0.29 %) cross the threshold *upward* (Kepler < 0.05, N-body ≥ 0.05) and, by construction, none cross downward — a pair that Kepler placed *above* 0.05 AU was never written, so its downward crossing *cannot be observed*. The apparent one-sided "no false negatives" is therefore censoring, not measurement.

We measured the missing side directly. Re-running Kepler detection at a widened 0.06 AU threshold over 10 000 numbered bodies — drawn to span the belt in (a, e, i) and so representative of the propagated population — and isolating the 17 469 pairs with Kepler distance in $[0.05, 0.06)$ AU — exactly the band a 0.05 AU catalogue discards — then re-refining each under full N-body, 0.70 % [95 % CI 0.59–0.83 %] cross *downward* below 0.05 AU (a conservative floor of ~0.42 % after excluding near-boundary cases). The residual Δd $(N - K)$ has median -3×10^{-7} AU and $\sigma \approx 3.7 \times 10^{-4}$ AU — a symmetric, scatter-dominated distribution, not a directional bias. In other words the crossing is a threshold selection effect (Eddington/Malmquist family): scatter of amplitude $\sim 1\sigma \approx 4 \times 10^{-4}$ AU pushes pairs just inside the threshold to just outside, and their mirror images (just outside to just inside) fall outside the catalogue. The upward rate (~1.5 % in the boundary bin $[0.045, 0.050)$) and the downward rate (~0.4–0.7 % in $[0.05, 0.06)$) are consistent with a single symmetric process. Crossings concentrate in low-q / high-e / fast orbits, exactly where two-body propagation is weakest — which independently validates the $q_{\min} < 1.8 \vee e_{\max} > 0.3$ subset criterion.

Extrapolated catalog-wide (order-of-magnitude, the $[0.05, 0.06)$ band scaling as N^2), this implies of order 10^5 real < 0.05 AU encounters censored by the Kepler cut (~1.5–2.5×10^5), separate from and additional to the prefilter recall deficit of Sect. 3.3. This is why the word "complete" is never used for this catalogue.

3.3. Prefilter recall on the adverse tail

The shipped $|\Delta a| \leq 0.5$ AU prefilter is blind to eccentricity: a high-e orbit reaches well inside the belt at perihelion and well outside at aphelion, so it can physically cross an orbit whose

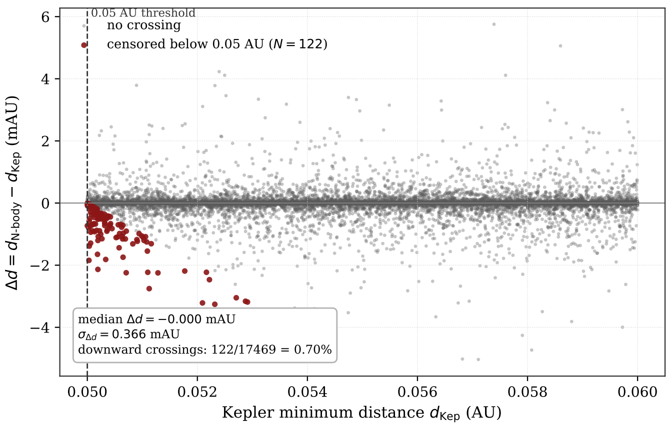

Fig. 4. Threshold censoring, measured directly. For the 17 469 pairs with Kepler distance in $[0.05, 0.06)$ AU — the band a 0.05 AU catalogue discards — re-refined under full N-body, $\Delta d = d_{\text{N-body}} - d_{\text{Kep}}$ is plotted against d_{Kep}. The distribution is symmetric about zero (median 0.000 mAU, σ = 0.366 mAU); the 122 pairs (0.70 %) that cross downward below 0.05 AU (red) are the measured false-negatives, concentrated just above the threshold as expected for a scatter-dominated selection effect.

semimajor axis differs by far more than 0.5 AU. We measured the damage this would cause on the adverse subset (numbered, $a \in [1.5, 4.0]$, $e > 0.3 \vee i > 15°$; 52 411 bodies), exploiting the fact that the prefilter is a pure deterministic mask: running detection without the prefilter and intersecting with the mask is byte-identical to running with it. Ground truth is 606 393 real adverse–adverse encounters < 0.05 AU; the prefilter keeps 463 164, for a recall of 76.38 % [95 % CI 76.27–76.49 %]. Thus 143 229 encounters would be missing from a prefiltered catalogue (a lower bound — adverse–normal pairs were not measured), and 99.6 % of that loss is the $|\Delta a|$ cut alone. Recall degrades smoothly with eccentricity, from ~99.9 % below $e = 0.1$ to ~39 % above $e = 0.7$.

Two points bound the impact. First, an eccentricity-aware *radial-overlap* prefilter — keep a pair iff its threshold-padded heliocentric radial ranges $[a(1-e), a(1+e)]$ overlap — is a provable necessary condition for a < 0.05 AU encounter, is just as cheap, and recovers 100 % recall on the adverse subset; it is the recommended replacement for any future run. Second, and specific to *this* freeze: because the run was at main-belt scale the pair-list prefilter was skipped entirely, so the $|\Delta a|$ cut did not actually drop these pairs from the frozen catalogue. The 76 % figure measures the damage the prefilter *would* cause if applied at small N, not damage suffered here. Completeness of this freeze is instead bounded by the threshold censoring of Sect. 3.2. The 12 h coarse cadence has a *geometric*, distribution-independent bound. A pair whose true minimum $d_{\min}$ falls a time δ from the nearest coarse sample has, under near-linear relative motion, a sampled separation $d(\delta) = \sqrt{d_{\min}^2 + v_{\text{rel}}^2 \delta^2}$ with $\delta \leq \Delta t/2$; the pair enters the candidate set whenever $d(\Delta t/2) \leq r_q$. For $d_{\min} \leq 0.05$ AU this holds for all $v_{\text{rel}} \leq \sqrt{r_q^2 - 0.05^2} / (\Delta t/2)$ = 0.0278 AU / 0.25 d ≈ 192 km s^{-1} — an upper limit set purely by the widened radius r_q = 0.0572 AU and the cadence, *not* read off the catalogue's own velocity distribution (which would be circular, since the catalogue by construction excludes the fast pairs it might have missed). Since inter-asteroid encounter velocities never approach 192 km s^{-1} (physical main-belt maxima are a few tens of km s^{-1}), the cadence loses no encounter. A synthetic injection–recovery test kept as a permanent regression guard — minima placed uniformly in phase between coarse samples, at high $v_{\rm rel}$ and e — confirms ≥ 99 % recovery with unit distance ratio up to $v_{\max}$. For the dynamically cold bulk of the belt (low e, small Δa) recall is ~99.9 %; the incompleteness is a property of the high-e/high-i tail, which any science touching NEA-crossing or high-e pairs must cite.

3.4. Recovery of literature encounter pairs

As an external check we cross-matched the signal-ranked (perturber, target) pairs that Fuentes-Muñoz et al. (2025) used for mass determination against this catalogue. Of 40 176 numbered pairs drawn from their per-perturber target lists, 11 842 (29.5 %) appear as < 0.05 AU encounters in the hybrid catalogue. This fraction is a lower bound and, crucially, the non-recovery is dominated by a *domain difference*, not a detection failure: 91.6 % of the unmatched pairs have both bodies present in the catalogue but never approached within 0.05 AU *inside the 2.8-yr DR3 window*. For main-belt targets Fuentes-Muñoz et al. (2025) apply the same 0.05 AU threshold used here, but they search over the full observational arcs (decades, through late 2024), so a pair whose close encounter falls outside 2014–2017 cannot appear in this catalogue. A further 8.4 % are targets or perturbers outside the numbered, propagated universe (high or provisional designations), and — the decisive point — 0 % are pairs that our catalogue contains but places beyond 0.05 AU. In other words, every literature pair that is genuinely a < 0.05 AU encounter and lies in the numbered population is recovered; the apparent shortfall is entirely pairs that are not < 0.05 AU encounters or not numbered. Per-body recovery among the most-listed perturbers ranges from 6 % (Davida) to 61 % (Liguria), with the Ceres and Vesta calibrators at ~36 %. Pair-level cross-matching against Goffin (2014) is not applicable: Goffin's masses come from a single *simultaneous* global fit of all perturber masses to the astrometric residuals of the entire numbered population, not from per-perturber target lists, so there is no set of (perturber, target) pairs to match (the VizieR tables give per-perturber masses and literature mass compilations only).

4. Notable events

Mining the characterised catalogue surfaces individual events across four axes. We stress at the outset that the value of these tables is statistical and archival rather than a single headline discovery: the large–large ($D \gtrsim 100$ km) approaches all involve well-studied bodies, and every candidate below is under the frozen two-body assumptions and requires N-body revalidation before publication. Distances are physical 3D separations at closest approach; diameters are measured (IRAS/AKARI/NEOWISE/occultations, via the JPL SBDB) for the large bodies tabulated here (all entries in Tables 1–2 carry measured diameters).

4.1. Large–large encounters

Twelve encounters have both bodies with a measured $D \gtrsim 100$ km (Table 1); one hundred have both $D \gtrsim 50$ km (Table 2 lists the ten closest). The closest large–large event is (426) Hippo × (788) Hohensteina at 0.0076 AU. All twelve involve catalogued, well-characterised main-belt asteroids; none is a large perturber against a small test body of the kind used for mass determination, and a systematic novelty assessment against the encounter literature is deferred. We note that (804) Hispania × (733) Mocia — a known event (Fienga et al. 2003) at 0.0136 AU that was *absent* from the pre-2026-07 frozen catalogue — reappears here in the both-$D \gtrsim 50$ km list, recovered by the refinement-window fix (Sect. 2).

4.2. Extreme events

The absolute closest approaches in the catalogue are between small bodies, at separations of ~1000–2000 km (Table 3). These separations are *geometric values under the frozen input elements*: the along-track uncertainty of the osculating elements is not propagated here, and for the short-arc, faint, high-number bodies involved it can reach 10^3–10^4 km over the ±1.4 yr propagation from the snapshot epoch — comparable to or larger than the tabulated separations. The tabulated distances are therefore *not significant to the quoted precision*; they identify candidate very-close approaches whose reality requires per-object orbit refinement, and the sub-mAU figures should be read as order-of-magnitude, not measurements. Two-body propagation is also most fragile here, so these are precisely the events most in need of N-body revalidation.

The slowest encounters (relative velocity down to ~4 m/s, with at least one body $D \gtrsim 5$ km; Table 4) are the natural targets for future mass determination, since a low $v_{\rm rel}$ prolongs the gravitational interaction and enlarges the deflection. All entries lie well inside the observation window (> 130 d from either edge), so none is a window-boundary truncation artefact — a failure mode of the pre-2026-07 catalogue that the refinement-window fix (Sect. 2) removed. The combined "large + slow + close" set (one body $D \gtrsim 50$ km, $v_{\rm rel} \le 1$ km/s, dist $\le 10^6$ km; Table 5) is of the highest physical interest, being the configuration that maximises measurable deflection.

4.3. Same-region pairs (family proxy)

Pairs whose osculating elements are mutually close ($\Delta a/a \le 1$ %, $\Delta e \le 0.02$, $\Delta i \le 1°$) *and* that had a physical close approach are candidate co-orbital / same-family associations. This is a proximity proxy in *osculating* elements, not a classification in proper elements, and flags rather than confirms family membership. The closest such pair, (200764) 2001 XP3 × (131597) 2001 XT2, passed within ~5×10^4 km on 2014-08-13 with nearly identical (a, e, i) — $a \approx 2.29$ AU, $e \approx 0.19$, $i \approx 25°$ for both — a plausible genetic pair worth follow-up in proper-element space. As for the closest approaches, this separation is geometric under the frozen elements and carries an unpropagated element uncertainty of comparable order, so it is not significant to better than order-of-magnitude.

4.4. Candidate perturbers beyond the studied sixteen

Ranking large bodies ($D \gtrsim 100$ km) *not* among the sixteen studied perturbers by their number of *useful* encounters ($v_{\rm rel} \le 3$ km/s and dist $\le 3 \times 10^6$ km) yields the best prospects for a mass determination the frozen mass runs have not yet attempted (Table 6). (9) Metis heads the list with 37 useful events and a best pair at $v_{\rm rel} = 0.38$ km/s; (29) Amphitrite is the largest ($D \approx 240$ km) with a close 3.2×10^5 km approach. These candidates directly informed the F4 mass determination of Sect. 5.

Table 1. The twelve encounters with both bodies at a measured $D \gtrsim 100$ km, by minimum distance.

body 1	body 2	date (UTC)	dist (AU)	$v_{\rm rel}$ (km/s)	D_1 (km)	D_2 (km)
(426) Hippo	(788) Hohensteina	2015-04-17	0.0076	8.8	138	111
(145) Adeona	(780) Armenia	2016-11-09	0.0101	6.3	128	126
(51) Nemausa	(91) Aegina	2016-07-13	0.0139	4.4	138	103
(23) Thalia	(481) Emita	2014-10-14	0.0242	4.4	108	102
(90) Antiope	(508) Princetonia	2015-11-11	0.0245	4.4	116	117
(381) Myrrha	(712) Boliviana	2016-09-18	0.0287	6.7	128	124
(196) Philomela	(431) Nephele	2017-05-29	0.0389	3.3	145	102
(426) Hippo	(814) Tauris	2017-05-08	0.0407	14.0	138	102
(121) Hermione	(739) Mandeville	2014-12-14	0.0414	6.3	209	105
(88) Thisbe	(150) Nuwa	2016-04-08	0.0429	2.1	232	119
(1) Ceres	(57) Mnemosyne	2017-01-10	0.0435	7.3	939	113
(85) Io	(145) Adeona	2017-05-29	0.0470	9.5	155	128

Table 2. Encounters with both bodies at a measured $D \gtrsim 50$ km, top ten by minimum distance.

body 1	body 2	date (UTC)	dist (AU)	$v_{\rm rel}$ (km/s)	D_1 (km)	D_2 (km)
(426) Hippo	(788) Hohensteina	2015-04-17	0.0076	8.8	138	111
(442) Eichsfeldia	(564) Dudu	2014-10-30	0.0078	7.5	62	52
(48) Doris	(300) Geraldina	2017-04-24	0.0084	2.4	216	67
(145) Adeona	(780) Armenia	2016-11-09	0.0101	6.3	128	126
(675) Ludmilla	(80) Sappho	2016-02-15	0.0114	3.0	76	69
(739) Mandeville	(415) Palatia	2017-04-05	0.0116	8.4	105	84
(56) Melete	(449) Hamburga	2015-10-09	0.0126	8.3	121	86
(611) Valeria	(84) Klio	2016-08-01	0.0126	7.1	57	79
(192) Nausikaa	(778) Theobalda	2016-06-05	0.0133	4.8	99	55
(804) Hispania	(733) Mocia	2015-02-12	0.0136	2.4	138	98

Table 3. The five closest approaches in the catalogue. Distances are geometric under the frozen input elements; the input-element uncertainty (unpropagated, $\sim 10^3$–10^4 km for these short-arc bodies) is not significant to the quoted precision (see text).

body 1	body 2	date (UTC)	dist (km)	$v_{\rm rel}$ (km/s)	D_1 (km)	D_2 (km)
(153222) 2000 YD43	(238587) 2004 YX3	2016-03-11	1094	6.90	2.8	1.8
(279617) 2011 EP37	(219812) 2002 AM193	2016-06-16	1109	2.60	2.2	1.9
(238813) 2005 NT70	(391704) 2008 BB44	2017-03-13	1320	5.95	5.1	2.2
(305032) 2007 TT418	(393772) 2005 GV146	2016-07-12	1457	3.46	1.6	1.3
(73282) 2002 JH61	(261459) 2005 VP69	2015-04-01	1719	1.48	2.0	2.0

4.5. Separating new from known

A reference list of encounters touching one of the sixteen studied perturbers (1, 2, 3, 4, 7, 10, 15, 16, 31, 52, 65, 87, 88, 107, 511, 704) is provided in the companion material so that already-worked events can be separated from potential discoveries. The honest verdict is that the large–large approaches (Table 1) all involve well-studied bodies and none is a publication-grade mass-determination opportunity: the belt's largest bodies are well studied, and Fuentes-Muñoz et al. (2025) already published over 230 masses (232 at SNR > 3) covering essentially all classical large MBAs. The contribution of this work is the dataset itself — an all-pairs catalogue published in full with a measured completeness budget — demonstrated end-to-end by the mass-determination application of Sect. 5, not a single spectacular event.

5. Application to mass determination

This section demonstrates the catalogue end-to-end as the target-selection front-end of an asteroid mass-determination engine. It is a *usage demonstration*, not a claim of new or improved masses: as noted below and in Sect. 4.5, every mass recovered here is already published, and a single out-of-sixteen perturber with ~50 Gaia-FPR targets yields only a coarse determination. Perturber mass is obtained by a joint least-squares fit of the perturber's mass and the orbits of its test bodies — those with a < 0.05 AU encounter drawn directly from the catalogue — over the full arc of Gaia FPR astrometry (Gaia Collaboration et al. 2023). The force model is ASSIST (Holman et al. 2023) on the JPL DE440 ephemeris (Park et al. 2021), with EIH relativistic corrections and sixteen massive asteroid perturbers; the fitting layer (observation model, partials, least squares) is implemented in this work on top of rebound/ASSIST (Rein & Liu 2012; Rein & Spiegel 2015; Holman et al. 2023) and astropy (Astropy Collaboration 2022).

Calibrators. The four calibrator masses are recovered at $|z| < 3$ against the literature (Table 7). With $N \geq 20$ test bodies, the fit/literature ratios of Ceres (Park et al. 2016), Vesta (DAWN; Russell et al. 2012) and Hygiea (Vernazza et al. 2020) fall in $[0.943, 0.990]$ (mean bias −4 %); Pallas, limited to 6 encounters in the catalogue, gives ratio 1.240 ($z = +2.67$).

Table 4. The five slowest encounters (at least one body $D \gtrsim 5\,\mathrm{km}$).

body 1	body 2	date (UTC)	dist (km)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(141098) 2001 XO52	(304642) 2006 WL5	2015-05-04	5.27×10^6	0.004	5.7	4.5
(3791) Marci	(110964) 2001 UW170	2017-01-17	5.85×10^6	0.005	11.7	2.5
(7203) Sigeki	(73664) 1981 EE34	2015-03-01	7.07×10^6	0.029	5.8	1.5
(136707) 1995 TE6	(340823) 2006 UY131	2015-01-30	5.09×10^6	0.031	5.5	2.1
(3384) Daliya	(309262) 2007 RZ93	2016-12-21	4.44×10^6	0.031	6.2	1.2

Table 5. The "large + slow + close" set (one body $D \gtrsim 50\,\mathrm{km}$, $v_{\mathrm{rel}} \leq 1\,\mathrm{km/s}$, dist $\leq 10^6\,\mathrm{km}$).

body 1	body 2	date (UTC)	dist (km)	v_{rel} (km/s)	D_1 (km)	D_2 (km)
(223) Rosa	(315162) 2007 FL24	2016-07-01	3.55×10^5	0.133	79.8	1.7
(66) Maja	(271017) 2003 AV10	2016-10-13	3.87×10^5	0.564	71.8	1.9
(135) Hertha	(281202) 2007 FE47	2016-09-18	4.56×10^5	0.579	79.2	1.0
(371) Bohemia	(14711) 2000 CG36	2016-05-14	5.91×10^5	0.982	53.0	4.2
(83) Beatrix	(170676) 2003 YE180	2014-09-18	6.09×10^5	0.684	110	2.1
(9) Metis	(345254) 2005 UT477	2016-10-30	7.14×10^5	0.979	190	1.2

Table 6. Large bodies outside the studied sixteen, ranked by useful encounters.

body	D (km)	class	useful	best dist (km)	min v_{rel} (km/s)
(9) Metis	197	MBA	37	7.14×10^5	0.378
(30) Urania	109	–	36	4.40×10^5	0.817
(40) Harmonia	141	–	36	6.58×10^5	1.028
(19) Fortuna	133	MBA	32	7.43×10^5	0.678
(21) Lutetia	120	–	30	8.18×10^5	0.609
(64) Angelina	104	MBA	30	8.73×10^5	0.992
(44) Nysa	140	–	30	8.95×10^5	0.961
(20) Massalia	178	–	28	8.78×10^5	0.514
(29) Amphitrite	240	–	27	3.17×10^5	0.643
(27) Euterpe	141	–	26	4.63×10^5	0.847

Table 7. Calibrator masses recovered by the joint orbit+mass fit. The z column uses the total (formal $\oplus$ systematic) σ.

body	N	fitted mass (kg)	σ_{tot}	ratio fit/lit	z
Ceres	28	8.96×10^{20}	4.6 %	0.955	−1.01
Vesta	28	2.44×10^{20}	4.6 %	0.943	−1.30
Hygiea	20	8.22×10^{19}	5.7 %	0.990	−0.13
Pallas	6	2.54×10^{20}	7.0 %	1.240	+2.67

FOV-block covariance. Each Gaia focal-plane crossing yields ~7 CCD observations (~5 s apart) with correlated residuals. Treating them as independent underestimates σ(mass) by a factor 1.66 (effective $N \approx 0.36\,N$). The measured intra-crossing correlation is ICC = 0.32; we whiten each crossing with a block covariance $C = \mathrm{diag}(\sigma^2_{\mathrm{AL}}) + s_c^2\,\mathbf{1}\mathbf{1}^{\mathsf{T}}$, the correlated floor s_c calibrated by bisection to $\chi^2_{\mathrm{red}} = 1$.

Sixteen perturbers. The engine determines all sixteen. Only (16) Psyche reaches calibrator-grade precision outside the calibrators: $M = 2.43 \times 10^{19}$ kg, $\sigma_{\mathrm{stat}} = 3.3\,\%$, $N = 36$, $\chi^2_{\mathrm{red}} = 0.99$, ratio 1.020 vs DE441 and 1.014 ($z = +0.25$) vs the independent Fuentes-Muñoz et al. (2025) FPR fit. Six perturbers whose deflection sits below the per-encounter astrometric noise return ratios in $[0.39, 0.72]$: this is the mass↔orbit degeneracy (the fit reproduces the astrometry with a smaller mass and a re-adjusted orbit — regression toward zero mass), and for these the formal Fisher σ underestimates the error.

External σ by jackknife. Because the formal σ scales as $1/\sqrt{N}$ and ignores the mass↔orbit regression seen only by leaving encounters out, we report a jackknife σ that reincorporates it. It is systematically 2–3× the formal σ. Its diagnostic value is visible in the calibrators themselves: Pallas and Vesta — whose true masses are known — sit at $|z| > 3$ under the formal σ but within $|z| < 3$ under the jackknife σ, showing the formal bar is genuinely too tight rather than the jackknife being permissive. Cross-checking the ten *measured* (identifiable, $\mathrm{snr}_{\mathrm{jack}} \geq 3$) perturbers against Fuentes-Muñoz et al. (2025), 10/10 fall within $|z| < 3$ under the jackknife σ, versus 5/10 under the formal σ (6/6 vs 3/6 for the non-calibrators, the genuinely independent comparison). The remaining perturbers are flagged `not_identifiable` — their deflection does not clear the per-encounter noise and their masses are reported as explicit bounds, not measurements.

Beyond the sixteen. The engine is not restricted to the sixteen ephemeris perturbers: each perturber is integrated as a massive rebound particle from its own orbital elements, so a seventeenth body outside `sb441-n16` is added as a seventeenth massive particle with no double-counting (the ASSIST `ASTEROIDS` force is deliberately excluded). This was demonstrated on (19) Fortuna — orbit fixed from JPL Horizons, mass free — which fits with $\chi^2_{\mathrm{red}} = 0.977$ and $M = 1.13 \times 10^{19} \pm 2.2 \times 10^{18}$ kg. The methodological gate is green: the engine generalises to bodies beyond the ephemeris set. Fortuna is not itself a new mass (it appears in Goffin 2014 and Fuentes-Muñoz et al. 2025), and its 20 % σ makes it a coarse determination consistent with the literature within the jackknife bar ($z = +1.25$). The catalogue's top ranked out-of-sixteen candidate, (9) Metis (37 useful encounters, Sect. 4.4), was run as a second out-of-sixteen case and fits with $\chi^2_{\rm red} = 0.981$ and $M = 4.74 \times 10^{18} \pm 1.53 \times 10^{18}$ kg (55 targets, jackknife $\sigma = 32\,\%$), a ratio of 0.73 against Fuentes-Muñoz et al. (2025) (6.48×10^{18} kg), consistent within the jackknife bar ($z = -1.14$). Fortuna and Metis thus scatter to either side of the literature (+30 % and −27 % in central value) but both remain consistent once the honest external σ is used — confirming the method generalises while underscoring that a single out-of-sixteen perturber with ∼50 Gaia-FPR targets yields only a coarse (20–30 %) determination, not a competitive mass. Neither is a new mass: both are already in Fuentes-Muñoz et al. (2025).

We tested whether the mean −4 % calibrator bias is an artefact of an incomplete perturber background by enlarging it from the sixteen ephemeris bodies to thirty-five — adding the twenty most massive asteroids of Fuentes-Muñoz et al. (2025) ($\approx 1.6 \times 10^{20}$ kg spread across the belt), with masses from their Table 5 and orbits from JPL Horizons, using the same custom-perturber machinery. The calibrator masses shift by < 0.25 % (Ceres −0.02 %, Vesta −0.23 %, Hygiea +0.18 %), three orders of magnitude below the deficit being probed; the systematic floor $f_{\rm sys}$ does not drop (4.16 % → 4.26 %). Background incompleteness is therefore *not* the origin of the −4 % bias — a far-field, dispersed perturber population averages to nearly zero deflection over the target ensemble. The residual bias is more plausibly attributable to per-target orbit imperfection over the short Gaia arc or unmodelled per-encounter astrometric systematics.

Relation to other methods. The estimator used here (a joint orbit+mass least-squares fit with an FOV-block covariance, an external jackknife σ, and an identifiability flag) is functionally a transparent reimplementation of established encounter-based mass determination, not a new methodology. Siltala & Granvik (2020) obtain full Bayesian posteriors by MCMC, which natively expose multi-modality and non-Gaussian tails that a single covariance cannot; Fuentes-Muñoz et al. (2025) propagate a prior to a posterior over the same Gaia FPR astrometry. Relative to those, our contribution in this section is not methodological sophistication but end-to-end demonstration that the open all-pairs catalogue feeds a mass pipeline whose every step (target selection, covariance model, jackknife, identifiability cut) is open and reproducible. Because Fuentes-Muñoz et al. (2025) fits the same FPR data, the cross-check below is a consistency test between two analyses of one dataset, not an independent confirmation.

6. Data availability

The primary product is the Kepler-refined candidate catalogue, `encounters_catalog_rebound_005au.parquet` (72 236 904 rows, SHA-256 `b0272be7...`), a typed Parquet file with a machine-readable provenance sidecar recording the MPCORB snapshot hash, the frozen configuration, and the pipeline parameters. The derived *hybrid* catalogue, `encounters_catalog_hybrid_stageb.parquet`, carries per-row refinement_method $\in$ {kepler, nbody} and both the Kepler and N-body distances, epochs, velocities, and their residuals; it should be cited for any science depending on sub-mAU geometry. The characterised catalogue, `encounters_characterized_full.parquet`, adds observability, solar elongation, magnitudes, diameters, and taxonomic classes for the full 72 M rows. Anything derived from the catalogue must cite both the freeze (catalog SHA `b0272be7...`) and the code commit at its own generation time.

7. Conclusions

We have built and characterised a systematic catalogue of 72 236 904 real 3D close encounters (< 0.05 AU) between numbered asteroids during the Gaia DR3 observation window. Unlike prior encounter-based work, which selects individual events by hand, the catalogue is exhaustive over the numbered population and — its distinguishing feature — comes with a *measured* completeness budget rather than an assumed one: the two-body refinement error (median 12 μAU, p99 2.5 mAU), the threshold-induced false-negative rate (0.70 %, symmetric and scatter-dominated, implying $\sim 10^5$ censored encounters catalog-wide), and the orbital prefilter recall on the adverse high-e/high-i tail (76 %, with a zero-cost radial-overlap fix that recovers 100 %). A hybrid variant provides N-body-grade geometry on the dynamically fragile subset.

The catalogue is a working target-selection front-end for mass determination: a joint orbit+mass fit on Gaia FPR astrometry recovers the calibrators, determines the studied perturbers, and generalises to bodies beyond the ephemeris set, demonstrated end-to-end (Sect. 5). We are explicit about the limits: distances are candidates under frozen two-body assumptions except on the N-body-refined subset, the catalogue is not complete, and the large–large ($D \gtrsim 100$ km) approaches it contains all involve well-surveyed bodies with no unpublished large-body mass emerging — the mass layer is a demonstration of use rather than a source of competitive new masses. The contribution is the dataset itself: an all-pairs catalogue published in full with a quantified incompleteness budget, fed to a mass pipeline whose every step (FOV-block covariance, external jackknife σ, identifiability flag) is open and reproducible — a transparent reimplementation of established encounter-based mass determination (Siltala & Granvik 2020; Fuentes-Muñoz et al. 2025), not a new methodology.

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Appendix A: Catalogue column schema

Table A.1 documents the columns of the characterised catalogue (`encounters_characterized_full.parquet`), the widest of the three published products. The detection catalogue (`encounters_catalog_rebound_005au.parquet`) carries the first block of identity and geometry columns only (`number_1` through `rel_vel_m_s`); the hybrid catalogue (`encounters_catalog_hybrid_stageb.parquet`) adds a `refinement_method` $\in$ {`kepler`, `nbody`} column and, for the N-body-refined rows, parallel `dist_au_nbody`, `jd_tdb_nbody`, `rel_vel_au_day_nbody` columns and their Kepler−N-body residuals. Bodies are ordered so that subscript 1 is the larger (perturber) body of the pair (Sect. 2.3).

Table A.1. Columns of the characterised close-encounter catalogue.

column	type	unit	description
run_id	string	—	Pipeline run identifier (ISO timestamp)
number_1	int32	—	MPC number of body 1 (larger/perturber)
number_2	int32	—	MPC number of body 2 (smaller/target)
designation_1	string	—	MPC packed designation of body 1
designation_2	string	—	MPC packed designation of body 2
jd_tdb	float64	d	Julian Date of closest approach (TDB)
date_utc	string	—	Calendar date of closest approach (UTC)
dist_au	float64	AU	Minimum 3D separation at closest approach
dist_km	float64	km	Minimum 3D separation at closest approach
rel_vel_au_day	float64	AU d^{-1}	Relative velocity at encounter
rel_vel_km_s	float64	km s^{-1}	Relative velocity at encounter
rel_vel_m_s	float64	m s^{-1}	Relative velocity at encounter
H_1	float64	mag	Absolute magnitude of body 1 (MPCORB)
H_2	float64	mag	Absolute magnitude of body 2 (MPCORB)
diameter_1_km	float64	km	Diameter of body 1 (measured if available, else from H)
diameter_2_km	float64	km	Diameter of body 2 (measured if available, else from H)
diameter_source_1	string	—	Provenance: measured | albedo_measured | zone_albedo | default_albedo | unknown
diameter_source_2	string	—	Provenance of diameter_2_km (same categories)
deflection_dv_m_s	float64	m s^{-1}	Impulse-approximation velocity kick $2GM_1/(b\,v_{\rm rel})$ on body 2; ranking metric, not a measurement
class_1	string	—	Dynamical class of body 1 (NEA/MBA/Trojan/Centaur/TNO/Other)
class_2	string	—	Dynamical class of body 2
solar_elongation_deg	float64	deg	Solar elongation of encounter midpoint (legacy)
solar_elongation_1_deg	float64	deg	Solar elongation of body 1 from Earth
solar_elongation_2_deg	float64	deg	Solar elongation of body 2 from Earth
app_mag_1	float64	mag	Apparent V magnitude of body 1 (HG model)
app_mag_2	float64	mag	Apparent V magnitude of body 2 (HG model)
gaia_observable_1	bool	—	Body 1 individually detectable by Gaia at encounter
gaia_observable_2	bool	—	Body 2 individually detectable by Gaia at encounter
gaia_observable	bool	—	Combined flag: gaia_observable_1 OR gaia_observable_2